

Project Title: Transforming EDAs to Dashboards – Google Play Store Apps Analysis

1. Dashboard Overview

The Google Play Store Apps Analysis Dashboard is an interactive business intelligence solution developed using Microsoft Power BI. The dashboard transforms the insights generated during Exploratory Data Analysis (EDA) into interactive visual reports that help stakeholders understand app performance, user engagement, installation trends, ratings, and customer sentiment.

The dashboard is divided into three analytical pages:

- **Executive Overview**
- **Rating Analysis**
- **User Sentiment Analysis**

The report allows users to interact with the data through slicers and dynamic visuals, enabling faster and data-driven decision-making.

2. Data Sources

The dashboard uses two datasets:

Google Play Store Apps Dataset

Contains application-related information such as:

- App Name
- Category
- Rating
- Reviews
- Installs
- Price
- Type (Free/Paid)
- Content Rating
- Genres
- Last Updated
- Android Version

User Reviews Dataset

Contains customer review information including:

- App Name
- Sentiment
- Sentiment Polarity
- Sentiment Subjectivity

3. Data Model

The dashboard follows a simple relational model.

Tables Used

PlayStore

Stores application information.

UserReviews

Stores customer review and sentiment information.

Relationship

PlayStore[App]



UserReviews[App]

The relationship between the two tables is established using the App column.

4. DAX Measures Used

The following DAX measures were created to calculate KPIs and support interactive visualizations.

Total Apps

Total Apps = COUNT(PlayStore[App])

Description

Calculates the total number of applications available in the dataset.

Total Categories

Total Categories = DISTINCTCOUNT(PlayStore[Category])

Description

Returns the total number of unique app categories.

Average Rating

Average Rating = AVERAGE(PlayStore[Rating])

Description

Calculates the average rating of all applications.

Total Installs

Total Installs = SUM(PlayStore[Installs])

Description

Calculates the total number of installations across all apps.

Free Apps

Free Apps =CALCULATE(COUNTROWS(PlayStore),PlayStore[Type]="Free")

Description

Counts all free applications available in the Play Store.

Paid Apps

Paid Apps =CALCULATE(COUNTROWS(PlayStore),PlayStore[Type]="Paid")

Description

Counts all paid applications.

Total Reviews

Total Reviews =SUM(PlayStore[Reviews])

Description

Calculates the total number of customer reviews.

Positive Reviews

Positive Reviews

=CALCULATE(COUNTROWS(UserReviews),UserReviews[Sentiment]="Positive")

Description

Counts the number of positive reviews.

Neutral Reviews

Neutral Reviews

=CALCULATE(COUNTROWS(UserReviews),UserReviews[Sentiment]="Neutral")

Description

Counts the number of neutral reviews.

Negative Reviews

Negative Reviews

=CALCULATE(COUNTROWS(UserReviews),UserReviews[Sentiment]="Negative")

Description

Counts the number of negative reviews.

Average Polarity

Average Polarity =AVERAGE(UserReviews[Sentiment_Polarity])

Description

Calculates the average sentiment polarity score to measure overall customer satisfaction.

5. Dashboard Page 1 – Executive Overview

Objective

The Executive Overview dashboard provides a high-level summary of the Google Play Store dataset through key performance indicators and category-level insights.

KPI Cards

The dashboard displays the following key metrics:

- Total Apps
- Total Categories
- Average Rating
- Total Installs
- Free Apps
- Paid Apps

These KPIs provide an instant overview of the Play Store ecosystem.

Visualizations

Top Categories by Installs (Horizontal Bar Chart)

Shows the categories with the highest installation counts.

Business Insight

Game, Family, and Tools categories attract the highest number of installations, indicating strong user demand.

Free vs Paid Apps (Donut Chart)

Compares the proportion of free and paid applications.

Business Insight

More than 90% of applications are free, highlighting the dominance of the freemium business model.

Category Distribution (Treemap)

Displays the number of applications available in each category.

Business Insight

The Family category contains the largest number of published applications.

Filters

- Category
- Type
- Content Rating

These slicers allow users to dynamically filter all visuals on the page.

6. Dashboard Page 2 – Rating Analysis

Objective

This page analyzes application ratings, review counts, and installation trends to understand user engagement and application quality.

Visualizations

Reviews vs Rating (Bubble Chart)

Analyzes the relationship between reviews, ratings, and installs.

Insight

Applications with higher review counts generally receive more installs and maintain higher ratings.

Top Categories by Average Rating (Bar Chart)

Displays the categories with the highest average ratings.

Insight

Medical, Events, and Education apps maintain relatively higher customer ratings.

Top 5 Apps by Reviews (Column Chart)

Shows the applications with the highest customer engagement.

Insight

Popular applications receive significantly more reviews than other apps.

Average Rating Trend (Area Chart)

Shows how average ratings have changed over time.

Insight

The average app rating has remained consistently above 4, indicating stable application quality.

Filters

- Category
 - App Name
 - Last Updated Year
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7. Dashboard Page 3 – User Sentiment Analysis

Objective

This dashboard evaluates customer feedback through sentiment analysis.

KPI Cards

- Positive Reviews
- Neutral Reviews
- Negative Reviews
- Average Polarity

These KPIs summarize overall customer satisfaction.

Visualizations

Sentiment Distribution (Pie Chart)

Displays the percentage of Positive, Neutral, and Negative reviews.

Insight

Positive reviews account for the largest share of customer feedback.

Top Categories with Positive Reviews (Horizontal Bar Chart)

Shows categories receiving the highest number of positive reviews.

Insight

Game and Family applications receive the greatest amount of positive feedback.

Polarity vs Subjectivity (Scatter Plot)

Examines the relationship between review polarity and subjectivity.

Insight

Most reviews are moderately positive with balanced subjectivity scores.

Category-wise Sentiment Analysis (Stacked Column Chart)

Compares positive, neutral, and negative reviews across app categories.

Insight

Most categories receive considerably more positive reviews than negative ones.

Filters

- Category
 - Sentiment
-

8. Interactive Features

The dashboard includes interactive slicers that enable users to analyze data dynamically.

Available filters include:

- Category
- Type
- App Name
- Content Rating

- Year
- Sentiment

Cross-filtering allows users to drill down into specific app categories and compare business performance.

9. Business Insights

The dashboard provides several valuable business insights:

- Free applications dominate the Google Play Store ecosystem.
 - Game and Family categories generate the highest user engagement.
 - Higher review counts generally correspond to higher installation numbers.
 - Average application ratings remain above 4, indicating good overall app quality.
 - Positive customer sentiment significantly exceeds negative sentiment.
 - Customer reviews provide valuable feedback that developers can use to improve application quality.
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10. Business Value

This dashboard helps stakeholders:

- Identify high-performing app categories.
 - Analyze customer satisfaction through sentiment analysis.
 - Compare free and paid application performance.
 - Understand installation trends.
 - Monitor application ratings.
 - Make data-driven product and marketing decisions.
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11. Tools & Technologies Used

- Power BI Desktop – Dashboard Development
- Power Query – Data Cleaning & Transformation
- DAX (Data Analysis Expressions) – KPI and Measure Creation
- Python (Pandas, NumPy, Matplotlib, Seaborn) – Exploratory Data Analysis
- Google Colab – Data Preprocessing & EDA
- CSV Files – Data Source

Interview Question

1. Why did you choose these chart types?

Answer:

I selected each visualization based on the type of data and the business question I wanted to answer.

- **KPI Cards** were used to highlight important metrics such as Total Apps, Total Installs, Average Rating, Free Apps, and Paid Apps because they provide an instant overview of the dataset.
- **Bar Charts** were used to compare app categories based on installs, ratings, and reviews since bar charts are ideal for comparing values across multiple categories.
- **Treemap** was chosen to visualize the distribution of applications across different categories. It effectively shows the proportion of apps in each category.
- **Donut Chart** was used to compare Free and Paid applications because it clearly represents part-to-whole relationships.
- **Scatter Plot** was selected to analyze the relationship between reviews, ratings, and installs, helping identify trends and outliers.
- **Area Chart** was used to display the average rating trend over time, making it easier to observe changes across years.
- **Pie Chart** was used in the Sentiment Analysis dashboard to show the proportion of Positive, Neutral, and Negative reviews.

I chose these chart types because they communicate insights clearly and allow stakeholders to interpret the data quickly.

2. Which insight was the most actionable and why?

Answer:

The most actionable insight was that **free applications dominate the Google Play Store and generate significantly higher installation numbers than paid applications.**

This insight is valuable because it suggests that developers should consider adopting a **freemium business model**, where users can download the app for free while revenue is generated through advertisements, subscriptions, or in-app purchases.

Another important finding was that **Game and Family categories have the highest number of installs and positive user reviews.** Businesses planning to launch new applications can use this information to identify high-demand categories and develop competitive products.

These insights directly support business decisions related to pricing strategy, product development, and market positioning.

3. How did you handle missing or inconsistent data?

Answer:

Data cleaning was one of the most important stages of the project.

I performed the following preprocessing steps:

- Identified missing values using Pandas functions such as `isnull()` and `info()`.
- Removed duplicate records to avoid biased analysis.
- Converted the **Installs** column from text to numeric format by removing commas and the "+" symbol.
- Converted the **Price** column by removing the "\$" symbol and changing the data type to numeric.
- Converted the **Reviews** column into integer format.
- Handled missing values in the **Rating** column by removing records with insufficient information where appropriate.
- Standardized category names and corrected inconsistent values.
- Converted the **Last Updated** column into a proper date format.

These steps ensured that the dataset was clean, consistent, and suitable for accurate analysis and visualization.

4. What further analysis would you perform next?

Answer:

If I continue working on this project, I would perform more advanced analyses, including:

- Analyzing how app size affects ratings and installations.
- Studying the impact of app updates on user ratings over time.
- Performing category-wise revenue analysis for paid applications.
- Conducting detailed sentiment analysis using Natural Language Processing (NLP) to identify frequently mentioned positive and negative keywords in user reviews.
- Building customer segmentation based on app categories and user behavior.
- Creating time-series analysis to identify seasonal installation trends.
- Comparing app performance across different Android versions.

These analyses would provide deeper insights into user preferences and application performance.

5. How would you extend this project into Machine Learning or Dashboard Development?

Answer:

This project can be extended in multiple ways.

Machine Learning

I would develop predictive models to solve business problems such as:

- Predicting app ratings using features like reviews, installs, category, and price.
- Predicting the expected number of installs for newly launched applications.

- Building a sentiment classification model to automatically classify new customer reviews as Positive, Neutral, or Negative.
- Developing a recommendation system to suggest similar applications based on user preferences and app categories.

Dashboard Development

The dashboard can also be enhanced by:

- Publishing it to the **Power BI Service** for online access and collaboration.
- Implementing Row-Level Security (RLS) for secure data access.
- Adding drill-through pages for detailed app-level analysis.
- Including advanced DAX measures such as Year-over-Year (YoY) growth, ranking, and dynamic KPIs.
- Integrating real-time data using APIs to monitor app performance continuously.
- Designing mobile-friendly dashboard layouts for better accessibility.

These enhancements would make the project more scalable, interactive, and valuable for business stakeholders.